

Travel Itinerary Planner

Problem Statement: Uses a given destination and budget to generate a detailed day-by-day itinerary covering travel, accommodation, and activities within budget.

The idea is to develop a personalized travel plans based on the user inputs like: destination and budget; I have also added other optional conditions like source, number of travellers, days and food preferences. The system was designed to be more flexible to the user input.

I started by first giving a prompt to gemini AI to get insights about how prompt-based reasoning and response was generated.

1. Dataset collection:

- Explored several websites and found most of them weren't accessible.
- Following Kaggle datasets were used as the static dataset:
 - Tourist destinations and attraction,
 - City ratings and climatic information,
 - Tourism dataset.
- These static datasets provide structured, reliable and baseline knowledge and reduces the dependency on APIs.

2. Data cleansing: All null values were removed; column names were standardised.

Embeddings and vectorDB

- Hugging Face embeddings were used to convert textual data to numerical vectors.
- The data were created into chunks before embedding, fixed size chunking was used.

- These embedded chunks were stored in vectorDB. This enables the semantic search instead of keyword matching.
- Due to restricted access to the websites, I used Wikipedia API to retrieve topic-based information. This is done to get the dynamic data.
- Web search tool DuckDuckGoSearchResults from the Lang chain community tool is used to retrieve the relevant URLs for the places, stay and food.

3. *LLM selection:*

- Initial approach was using Gemini API but then limit exceeded.
- Then tried GROQ platform and worked with models: Llama 3.3 70B Versatile, llama-3.1-8b-instant and upgraded to openai/gpt-oss-120b.

4. Session memory was implemented.

5. Initial UI was built using streamlit and then extended to html, css, js and fast API where fastAPI handles the API.

6. Challenges:

- Explored tavily and serp APIs - for search engine results.
- Used rapiapi key for flight details.

[github repo link](#)